

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** 2-n-Butoxyethanol

**CAS No** 111-76-2

**Synonyms** Butyl cellosolve; Ethylene glycol monobutyl ether, Butyl glycol

**Product Code** **A17976**

**Address** ThermoFisher Scientific Australia Pty Ltd  
5 Caribbean Drive, Scoresby  
VICTORIA 3179, Australia

**Emergency Tel.** **CHEMTREC®**  
**03 9757 4559 or +613 9757 4559**

**Telephone / Fax Numbers** Tel: 1300 735 292  
Fax: 1800 067 639

**E-mail address** ANZinfo@thermofisher.com

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

**Physical hazards**  
No hazards identified

**Health hazards**

|                                    |            |
|------------------------------------|------------|
| Acute Oral Toxicity                | Category 4 |
| Acute Inhalation Toxicity - Vapors | Category 3 |
| Skin Corrosion/Irritation          | Category 2 |
| Serious Eye Damage/Eye Irritation  | Category 2 |

**Environmental hazards**  
No hazards identified

### Label Elements



Skull and Crossbones

**Signal Word****Danger****Hazard Statements**

H302 - Harmful if swallowed  
H315 - Causes skin irritation  
H319 - Causes serious eye irritation  
H331 - Toxic if inhaled  
Combustible liquid

**Precautionary Statements**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P311 - Call a POISON CENTER or doctor  
P330 - Rinse mouth  
P362 + P364 - Take off contaminated clothing and wash it before reuse  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P405 - Store locked up  
P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

Toxic to terrestrial vertebrates  
This product does not contain any known or suspected endocrine disruptors

## Section 3 - Composition and Information on Ingredients

| Component       | CAS No   | Weight % |
|-----------------|----------|----------|
| 2-Butoxyethanol | 111-76-2 | <100     |

## Section 4 - First Aid Measures

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

**Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately.

|                                            |                                                                                                                                                                                            |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice. |
| <b>General Advice</b>                      | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                          |
| <b>Self-Protection of the First Aider</b>  | Use personal protective equipment as required.                                                                                                                                             |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                       |
| <b>Most important symptoms and effects</b> | None reasonably foreseeable. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting                                                                           |
| <b>Notes to Physician</b>                  | Treat symptomatically. Symptoms may be delayed.                                                                                                                                            |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

### Extinguishing media which must not be used for safety reasons

Do not use a solid water stream as it may scatter and spread fire.

### Hazardous Decomposition Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), peroxides.

### Specific Hazards Arising from the Chemical

Combustible material. Containers may explode when heated. Vapors may accumulate in confined areas (basement, tanks, hopper/tank cars, etc.). Vapors may form explosive mixtures with air. Keep product and empty container away from heat and sources of ignition. Risk of ignition.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6 - Accidental Release Measures

### Emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas. Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental Precautions

Should not be released into the environment.

### Methods for Containment and Clean Up

#### Clean-up methods - small spillage

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition.

#### Clean-up methods - large spillage

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition.

**Conditions for Safe Storage, Including any Incompatibilities**

Keep away from heat, sparks and flame. Protect from light. Protect from moisture. Reacts with air to form peroxides. Keep containers tightly closed in a cool, well-ventilated place. Keep in properly labeled containers. Keep away from heat. Flammables area.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## Section 8 - Exposure Controls and Personal Protection

**Exposure limits**

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

**ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

**UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

**DE** - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

**NZ** - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

| Component       | Australia                                                                                 | New Zealand WEL                                   | ACGIH TLV   | The United Kingdom                                                                                                       | Germany                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------|-------------------------------------------------------------------------------------------|---------------------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2-Butoxyethanol | STEL: 50 ppm<br>STEL: 242 mg/m <sup>3</sup><br>TWA: 20 ppm<br>TWA: 96.9 mg/m <sup>3</sup> | TWA: 25 ppm<br>TWA: 121 mg/m <sup>3</sup><br>Skin | TWA: 20 ppm | STEL: 50 ppm 15 min<br>STEL: 246 mg/m <sup>3</sup> 15 min<br>TWA: 25 ppm 8 hr<br>TWA: 123 mg/m <sup>3</sup> 8 hr<br>Skin | TWA: 10 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 49 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 10 ppm (8 Stunden). MAK applies for the sum of the concentrations of 2-Butoxyethanol and its Acetate in air<br>TWA: 49 mg/m <sup>3</sup> (8 Stunden). MAK applies for the sum of the concentrations of 2-Butoxyethanol and its Acetate in air<br>Höhepunkt: 20 ppm<br>Höhepunkt: 98 mg/m <sup>3</sup><br>Haut |

**Biological limit values**

**UK** - Biological Monitoring Guidance Values provided by the UK's Health and Safety Executive (HSE) Control of Substances Hazardous to Health Regulations (COSHH) 2002 (as amended) and EH40/2005.

| Component       | Australia | New Zealand | European Union | United Kingdom                                              | Germany                                                                                      |
|-----------------|-----------|-------------|----------------|-------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| 2-Butoxyethanol |           |             |                | Butoxyacetic acid: 240 mmol/mol creatinine urine post shift | Butoxyacetic acid (after hydrolysis): 150 mg/g Creatinine urine (for long-term exposures: at |

|  |  |  |  |  |                                                                                                                                   |
|--|--|--|--|--|-----------------------------------------------------------------------------------------------------------------------------------|
|  |  |  |  |  | the end of the shift after several shifts )<br>Butoxyacetic acid (after hydrolysis): 150 mg/g<br>Creatinine urine (end of shift ) |
|--|--|--|--|--|-----------------------------------------------------------------------------------------------------------------------------------|

**Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material  | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments                                                                 |
|-----------------|-------------------|-----------------|-----------------|--------------------------------------------------------------------------------|
| Butyl rubber    | > 480 minutes     | 0.5 mm          | AS/NZS 2161     | As tested under EN374-3 Determination of Resistance to Permeation by Chemicals |
| Viton (R)       | > 480 minutes     | 0.4 mm          |                 |                                                                                |
| Nitrile rubber  |                   |                 |                 |                                                                                |
| Neoprene gloves | > 480 minutes     | 0.45 mm         |                 |                                                                                |
| Nitrile rubber  | > 480 minutes     | 0.56 mm         |                 |                                                                                |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Long sleeved clothing

**Respiratory Protection**

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:**

Organic gases and vapours filter Type A Brown conforming to EN14387 (or AUS/NZ equivalent)

**Recommended half mask:-**

Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures**

When using do not eat, drink or smoke. Provide regular cleaning of equipment, work area and clothing.

**Environmental exposure controls**

No information available.

## Section 9 - Physical and Chemical Properties

**Information on basic physical and chemical properties****Appearance**

Light yellow

**Physical State**

Liquid

**Odor**

Slight

**Odor Threshold**

No data available

**pH**

No information available

|                                                |                                                 |                                                                                                                    |
|------------------------------------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Melting Point/Range</b>                     | -70 °C / -94 °F                                 |                                                                                                                    |
| <b>Softening Point</b>                         | No data available                               |                                                                                                                    |
| <b>Boiling Point/Range</b>                     | 171 °C / 339.8 °F                               |                                                                                                                    |
| <b>Flash Point</b>                             | 62 °C / 143.6 °F                                | <b>Method</b> - Pensky Martens Closed Cup (ASTM D93, BS EN 22719, BS 2000 Part 404, IP 404, ISO 2719, AS/NZS 2106) |
| <b>Evaporation Rate</b>                        | No data available                               |                                                                                                                    |
| <b>Flammability (solid,gas)</b>                | Not applicable                                  | Liquid                                                                                                             |
| <b>Explosion Limits</b>                        | <b>Lower</b> 1.1 vol%<br><b>Upper</b> 10.6 vol% |                                                                                                                    |
| <b>Vapor Pressure</b>                          | 0.8 hPa @ 20°C                                  |                                                                                                                    |
| <b>Vapor Density</b>                           | No data available                               | (Air = 1.0)                                                                                                        |
| <b>Specific Gravity / Density</b>              | 0.901                                           | @ 20 °C                                                                                                            |
| <b>Bulk Density</b>                            | Not applicable                                  | Liquid                                                                                                             |
| <b>Water Solubility</b>                        | Miscible                                        |                                                                                                                    |
| <b>Solubility in other solvents</b>            | No information available                        |                                                                                                                    |
| <b>Partition Coefficient (n-octanol/water)</b> |                                                 |                                                                                                                    |
| <b>Component</b>                               | <b>log Pow</b>                                  |                                                                                                                    |
| 2-Butoxyethanol                                | 0.81                                            |                                                                                                                    |
| <b>Autoignition Temperature</b>                | 230 °C / 446 °F                                 |                                                                                                                    |
| <b>Decomposition Temperature</b>               | No data available                               |                                                                                                                    |
| <b>Viscosity</b>                               | 5.31 mPa.s at 20 °C                             | Dynamic                                                                                                            |
| <b>Explosive Properties</b>                    |                                                 | explosive air/vapour mixtures possible                                                                             |
| <b>Oxidizing Properties</b>                    | Not oxidising                                   |                                                                                                                    |
| <b>Other information</b>                       |                                                 |                                                                                                                    |
| <b>Molecular Formula</b>                       | C6 H14 O2                                       |                                                                                                                    |
| <b>Molecular Weight</b>                        | 118.18                                          |                                                                                                                    |

## Section 10 - Stability and Reactivity

|                                         |                                                                                                                                                                                                           |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reactivity</b>                       | None known, based on information available                                                                                                                                                                |
| <b>Stability</b>                        | Stable under normal conditions.                                                                                                                                                                           |
| <b>Conditions to Avoid</b>              | Incompatible products, Keep away from open flames, hot surfaces and sources of ignition, Extremes of temperature and direct sunlight, Exposure to air or moisture over prolonged periods, Heating in air. |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Bases, Metals, Aluminium, . Zinc                                                                                                                                                 |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). peroxides.                                                                                                                                       |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                                                                                                                  |

## Section 11 - Toxicological Information

### Information on Toxicological Effects

#### Product Information

|                            |                                                                  |
|----------------------------|------------------------------------------------------------------|
| <b>(a) acute toxicity;</b> |                                                                  |
| <b>Oral</b>                | Category 4                                                       |
| <b>Dermal</b>              | Based on available data, the classification criteria are not met |
| <b>Inhalation</b>          | Category 3                                                       |

| Component       | LD50 Oral          | LD50 Dermal                    | LC50 Inhalation            |
|-----------------|--------------------|--------------------------------|----------------------------|
| 2-Butoxyethanol | 1746 mg/kg ( Rat ) | LD50 > 2000 mg/kg (Guinea pig) | LC50 = 450 ppm ( Rat ) 4 h |

|  |  |          |                            |
|--|--|----------|----------------------------|
|  |  | OCED 402 | LC50 = 486 ppm ( Rat ) 4 h |
|--|--|----------|----------------------------|

(b) skin corrosion/irritation; Category 2  
 Test method OECD 405  
 Test species rabbit  
 Observational endpoint Irritating to skin

(c) serious eye damage/irritation; Category 2

(d) respiratory or skin sensitization;

Respiratory

Based on available data, the classification criteria are not met

Skin

Based on available data, the classification criteria are not met

| Component                            | Test method                            | Test species | Study result      |
|--------------------------------------|----------------------------------------|--------------|-------------------|
| 2-Butoxyethanol<br>111-76-2 ( <100 ) | Guinea Pig Maximisation Test<br>(GPMT) | guinea pig   | - non-sensitising |

(e) germ cell mutagenicity; Based on available data, the classification criteria are not met

(f) carcinogenicity; Based on available data, the classification criteria are not met  
 The table below indicates whether each agency has listed any ingredient as a carcinogen

(g) reproductive toxicity; Based on available data, the classification criteria are not met  
 Reproductive Effects None known

(h) STOT-single exposure; Based on available data, the classification criteria are not met

(i) STOT-repeated exposure; Based on available data, the classification criteria are not met

Target Organs

None known.

(j) aspiration hazard; Based on available data, the classification criteria are not met

Other Adverse Effects No information available

Symptoms / effects, both acute and delayed Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

## Section 12 - Ecological Information

Ecotoxicity effects . Do not empty into drains.

| Component       | Freshwater Fish                            | Water Flea                                                                    | Freshwater Algae     | Microtox |
|-----------------|--------------------------------------------|-------------------------------------------------------------------------------|----------------------|----------|
| 2-Butoxyethanol | 1490 mg/L LC50 96 h<br>2950 mg/L LC50 96 h | 1550 mg/l EC50 48 hr<br>>1000 mg/L EC50 48 h<br>1698 - 1940 mg/L EC50<br>24 h | 1840 mg/l EC50 72 hr |          |

Persistence and Degradability Readily biodegradable  
 Persistence Persistence is unlikely.

| Component                            | Degradability       |
|--------------------------------------|---------------------|
| 2-Butoxyethanol<br>111-76-2 ( <100 ) | 90% (28d) OECD 301B |

Bioaccumulative Potential Bioaccumulation is unlikely

| Component       | log Pow | Bioconcentration factor (BCF) |
|-----------------|---------|-------------------------------|
| 2-Butoxyethanol | 0.81    | No data available             |

Mobility The product is water soluble, and may spread in water systems. : Will likely be mobile in the environment due to its water solubility Spillage unlikely to penetrate soil: Highly mobile in soils

|                                        |                                                                           |
|----------------------------------------|---------------------------------------------------------------------------|
| <b>Endocrine Disruptor Information</b> | This product does not contain any known or suspected endocrine disruptors |
| <b>Persistent Organic Pollutant</b>    | This product does not contain any known or suspected substance            |
| <b>Ozone Depletion Potential</b>       | This product does not contain any known or suspected substance            |

## Section 13 - Disposal Considerations

|                                            |                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations. |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point.                                                                                                                                                                                                                                            |
| <b>Other Information</b>                   | Chemical wastes should be disposed through a licensed commercial waste collection service. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.                                                                                             |

## Section 14 - Transport Information

### IMDG/IMO

|                                |                               |
|--------------------------------|-------------------------------|
| <b>UN-No</b>                   | UN2810                        |
| <b>Proper Shipping Name</b>    | TOXIC LIQUID, ORGANIC, N.O.S. |
| <b>Technical Shipping Name</b> | 2-Butoxyethanol               |
| <b>Hazard Class</b>            | 6.1                           |
| <b>Packing Group</b>           | III                           |

### ADG

|                                |                               |
|--------------------------------|-------------------------------|
| <b>UN-No</b>                   | UN2810                        |
| <b>Proper Shipping Name</b>    | TOXIC LIQUID, ORGANIC, N.O.S. |
| <b>Technical Shipping Name</b> | 2-Butoxyethanol               |
| <b>Hazard Class</b>            | 6.1                           |
| <b>Packing Group</b>           | III                           |

### IATA

|                                |                               |
|--------------------------------|-------------------------------|
| <b>UN-No</b>                   | UN2810                        |
| <b>Proper Shipping Name</b>    | TOXIC LIQUID, ORGANIC, N.O.S. |
| <b>Technical Shipping Name</b> | 2-Butoxyethanol               |
| <b>Hazard Class</b>            | 6.1                           |
| <b>Packing Group</b>           | III                           |

|                               |                                 |
|-------------------------------|---------------------------------|
| <b>Environmental hazards</b>  | No hazards identified           |
| <b>Special Precautions</b>    | No special precautions required |
| <b>Additional information</b> | None known                      |

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

|                                    |                  |
|------------------------------------|------------------|
| <b><u>National Regulations</u></b> | <b>Australia</b> |
|------------------------------------|------------------|

See section 8 for national exposure control parameters.



**Standard for the Uniform Scheduling of Medicines and Poisons**

Classified as a scheduled poison according to the Standard for Uniform Scheduling of Medicines and Poisons.

| Component                  | Standard for the Uniform Scheduling of Medicines and Poisons                                                                                                                                                                                                                                     |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2-Butoxyethanol - 111-76-2 | Schedule 6 listed - except in plant growth regulator preparations containing ≤20% of such substances, or in other preparations containing ≤10% of such substances Schedule 6 listed - except when separately specified in these Schedules, or in preparations containing ≤10% of such substances |

**Australian Industrial Chemicals Introduction Scheme (AICIS)**

| Component                  | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information                                                                                                                                                                                                       |
|----------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2-Butoxyethanol - 111-76-2 | Present                                                     | Specific information requirement: Obligations to provide information apply. You must tell us within 28 days if the circumstances of your importation or manufacture (introduction) are different to those in our assessment. |

**Australian - Illicit Drug Precursors/Reagents Substance List**

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

**Chemicals of Security Concern**

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

**National pollutant inventory** Not applicable

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

**International Inventories**

| Component       | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|-----------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| 2-Butoxyethanol | X    | X     | 203-905-0 | -      | X    | X   | -    | X     | X    | X    | X     | KE-04134 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**Basel convention on the control of transboundary movements of hazardous wastes and their disposal**

Not applicable.

| Component       | CAS No   | OECD HPV | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------------|----------|----------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| 2-Butoxyethanol | 111-76-2 | Listed   | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

**Authorisation/Restrictions according to EU REACH**

| Component       | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 2-Butoxyethanol | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

## Section 16 - Other Information

**Legend**

|                                                                                                      |                                                                                                                              |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>AICS</b> - Australian Inventory of Chemical Substances                                            | <b>NZIoC</b> - New Zealand Inventory of Chemicals                                                                            |
| <b>TSCA</b> - United States Toxic Substances Control Act Section 8(b) Inventory                      | <b>EINECS/ELINCS</b> - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances |
| <b>DSL/NDL</b> - Canadian Domestic Substances List/Non-Domestic Substances List                      | <b>ENCS</b> - Japanese Existing and New Chemical Substances                                                                  |
| <b>IECSC</b> - Chinese Inventory of Existing Chemical Substances                                     | <b>KECL</b> - Korean Existing and Evaluated Chemical Substances                                                              |
| <b>PICCS</b> - Philippines Inventory of Chemicals and Chemical Substances                            | <b>CAS</b> - Chemical Abstracts Service                                                                                      |
| <b>TWA</b> - Time Weighted Average                                                                   | <b>ACGIH</b> - American Conference of Governmental Industrial Hygienists                                                     |
| <b>IARC</b> - International Agency for Research on Cancer                                            | Predicted No Effect Concentration (PNEC)                                                                                     |
| <b>ICAO/IATA</b> - International Civil Aviation Organization/International Air Transport Association | <b>IMO/IMDG</b> - International Maritime Organization/International Maritime Dangerous Goods Code                            |
| <b>MARPOL</b> - International Convention for the Prevention of Pollution from Ships                  |                                                                                                                              |
| <b>NZS 5433:2020</b> - Transport of Dangerous Goods on Land                                          | <b>OECD</b> - Organisation for Economic Co-operation and Development                                                         |
| <b>LD50</b> - Lethal Dose 50%                                                                        | <b>LC50</b> - Lethal Concentration 50%                                                                                       |
| <b>EC50</b> - Effective Concentration 50%                                                            | <b>ATE</b> - Acute Toxicity Estimate                                                                                         |
| <b>WEL</b> - Workplace Exposure Limit                                                                | <b>RPE</b> - Respiratory Protective Equipment                                                                                |
| <b>DNEL</b> - Derived No Effect Level                                                                | <b>NOEC</b> - No Observed Effect Concentration                                                                               |
| <b>POW</b> - Partition coefficient Octanol:Water                                                     | <b>BCF</b> - Bioconcentration factor                                                                                         |
| <b>vPvB</b> - very Persistent, very Bioaccumulative                                                  | <b>PBT</b> - Persistent, Bioaccumulative, Toxic                                                                              |
| <b>VOC</b> - (Volatile Organic Compound)                                                             |                                                                                                                              |

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

Chemical incident response training.

**Revision Date** 11-Sep-2023  
**Revision Summary** SDS sections updated.

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of**

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**Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).****Disclaimer**

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**End of Safety Data Sheet**